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Automated Five-Stage Diabetic Retinopathy Grading Using a Leak-Safe and Explainable Deep-Handcrafted Fusion Framework

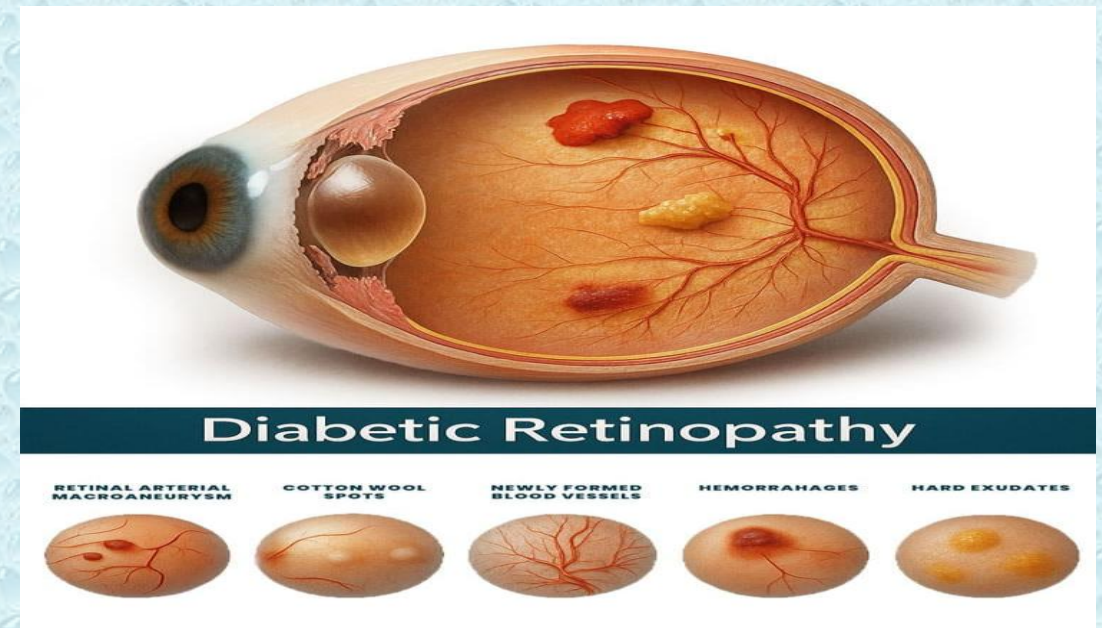
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Presentation Outline

Background, burden, and DR grading scale

Problem statement and research objectives

Dataset construction and leak-safe preparation

Proposed stage-wise framework and evaluation protocol

Results, explainability, and web-based prototype

Conclusion, limitations, and future work

Background and Motivation

- ❑ Diabetic retinopathy (DR) is a diabetes-related retinal microvascular disease and a major cause of preventable vision loss.
- ❑ Early-stage DR may progress silently, making routine fundus screening essential.
- ❑ Current screening depends on expert manual grading of fundus images, which is time-consuming and difficult to scale.
- ❑ Automated and explainable DR grading can support scalable screening in resource-limited settings.

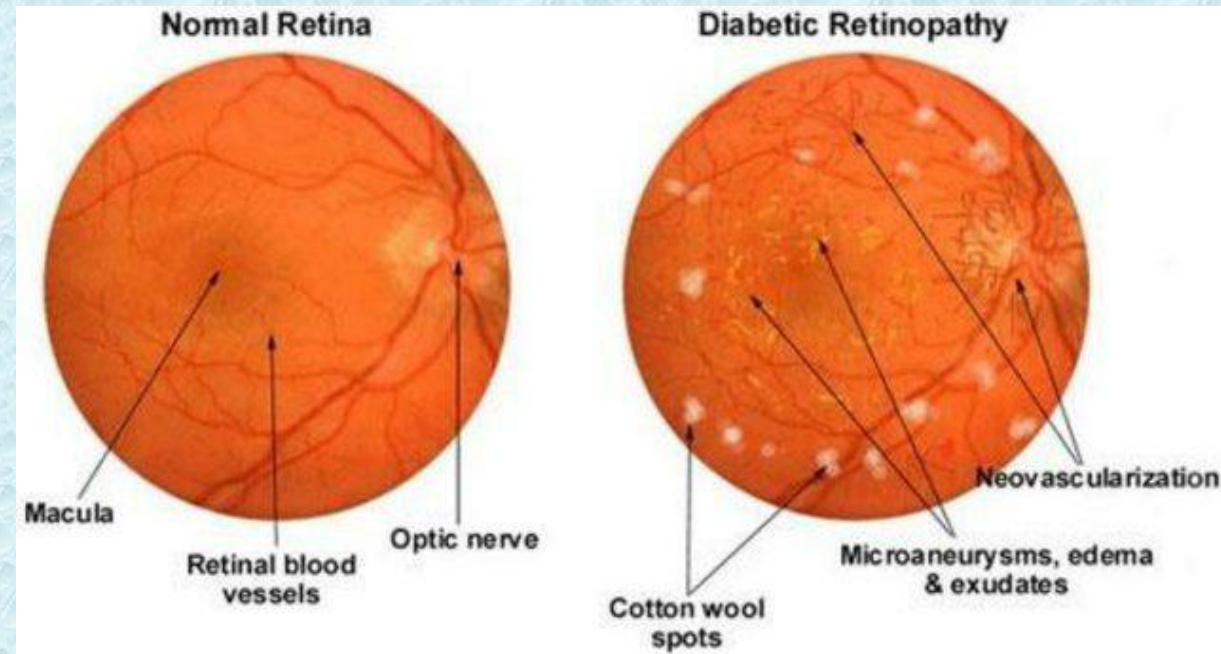


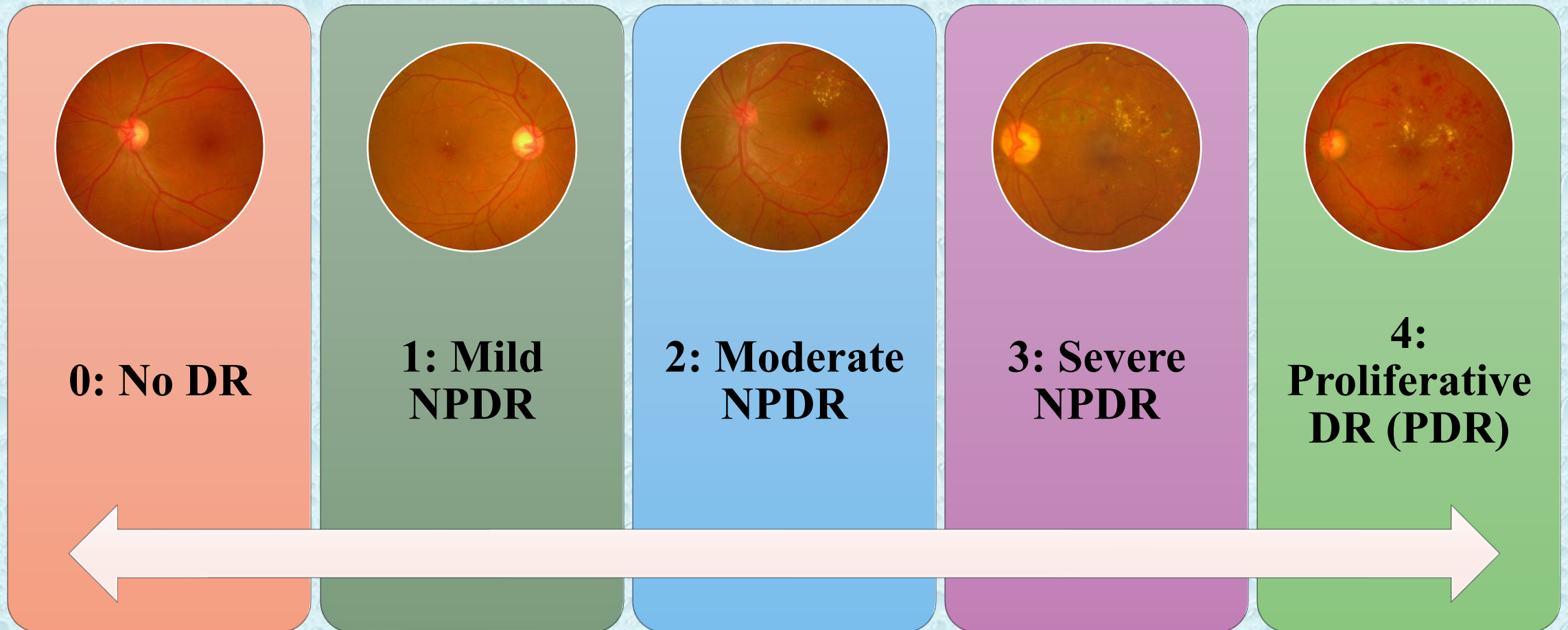
Figure 1: Normal retina versus diabetic retinopathy affected retina showing common lesions.

DR Burden: Global and Bangladesh Context

- **Global burden:** Diabetes is projected to affect approximately **693 million adults**, increasing the future risk of DR-related vision loss.
- **Bangladesh context:** More than **13 million** people are living with diabetes, with about **1.85 million** already at risk of DR-related vision loss.
- **Prevalence pattern:** Reported DR prevalence is around **18–20%**, rising to **>40%** after longer diabetes duration.
- **Screening Gap:** Low screening uptake and limited specialist availability can delay diagnosis and increase preventable blindness risk.

Five-Stage DR Severity Scale: Grades 0-4

Unified label space used for DR grading scale (0–4)



Five-Stage DR Severity Grading and Key Lesions

Grade	Label	Key Visual Cue
0	No DR	No clinically visible DR lesion
1	Mild NPDR	Few microaneurysms
2	Moderate NPDR	Increased microaneurysms, hemorrhages, and/or exudates
3	Severe NPDR	Widespread hemorrhages and ischemic changes
4	Proliferative DR (PDR)	Neovascularization and vision-threatening changes

Problem Statement and Research Need

- ❑ **Goal:** Automated five-stage DR grading from retinal fundus images, covering Grades 0–4.
- ❑ **Key challenges:** Expert-dependent grading, subtle lesion overlap, class imbalance, and resource-limited screening constraints.
- ❑ **Research gap:** Limited leak-safe cross-source evaluation under realistic mixed-source screening variability.
- ❑ **Need:** Explainable deep-handcrafted fusion framework with fixed Train/Validation/Test evaluation, bootstrap uncertainty, and Grad-CAM++ support.

Research Objectives

- ❑ Build a leak-safe APTOS 2019 + IDRiD benchmark for five-class DR grading.
- ❑ Develop a five-stage deep-handcrafted fusion framework with validation-guided component selection.
- ❑ Identify the strongest deep and handcrafted representations for fine-grained DR grading.
- ❑ Evaluate locked final-stage fusion candidates on a held-out test set with bootstrap uncertainty.
- ❑ Deploy the selected model in an explainable Grad-CAM++ web prototype for screening support.

Datasets Used in this Study

Item	Description
Data modality	Color retinal fundus photographs for five-class DR grading
Dataset 1	APTOS 2019 Blindness Detection: 3,662 original images → 3,534 used after duplicate screening
Dataset 2	IDRiD / ISBI-2018 Challenge Dataset: official split preserved; 408 train images used from 413 and 103 test images retained
Final dataset	4,045 unique images labeled into Grades 0–4
Why two datasets?	To increase imaging diversity and support realistic mixed-source evaluation

Class Distribution and Imbalance

Key takeaway: The final benchmark is class-imbalanced, with **Grade 0** and **Grade 2** dominating, while **Severe NPDR** and **PDR** are underrepresented. Therefore, evaluation should consider Macro-F1, Cohen's κ , OvR AUC, and bootstrap uncertainty in addition to Accuracy.

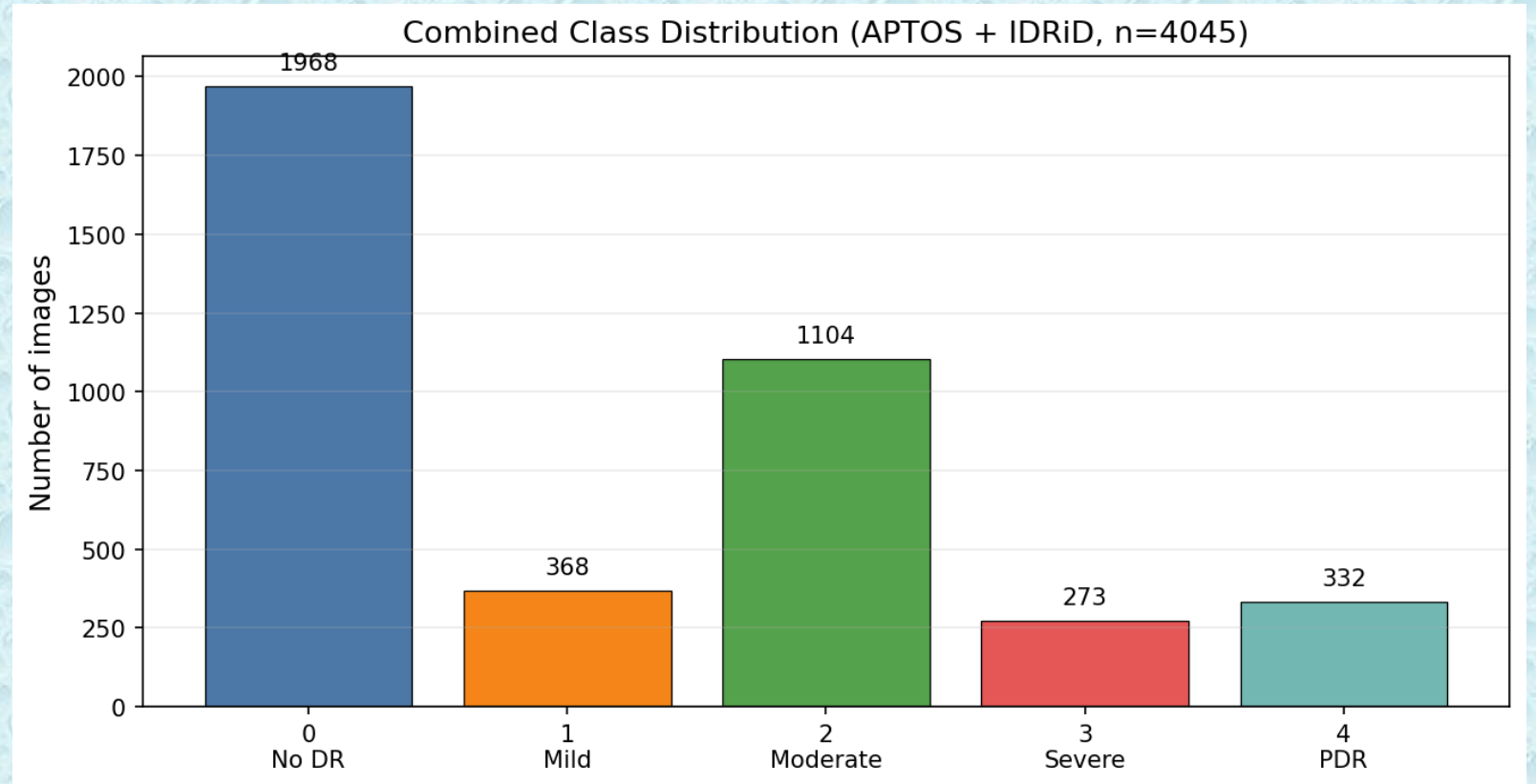


Figure 2: Combined class distribution of the final dataset (APTOS+IDRiD, n=4045).

Leak-Safe Split Strategy

- **APTOS 2019:** stratified **70/15/15** Train / Validation / Test split.
- **IDRiD:** official Test split preserved; Train split **85/15** into Train / Validation.
- **Blended benchmark:** APTOS and IDRiD merged **split-wise** to keep evaluation folds independent.
- **Final split:** Train = **2,819**, Validation = **592**, Test = **634**.

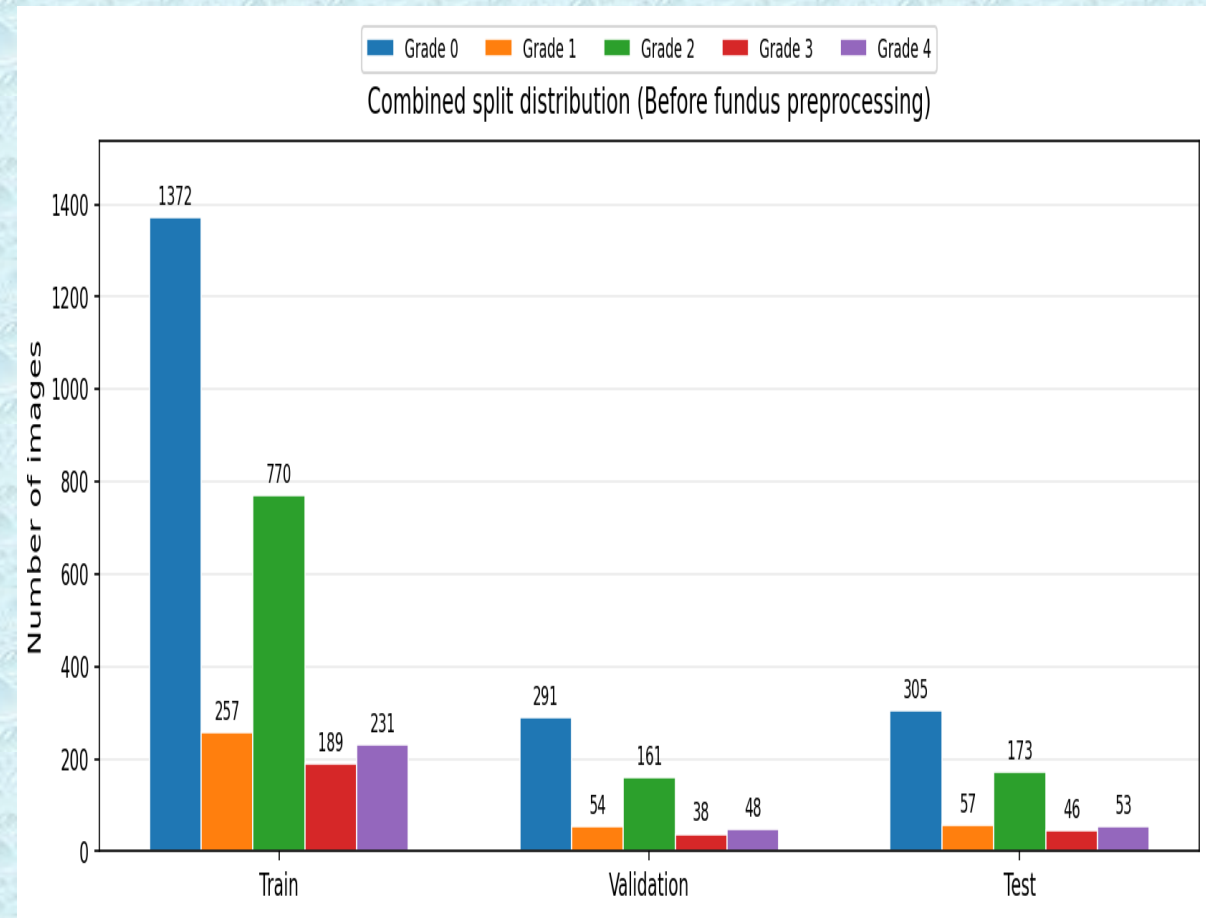


Figure 3: Split-wise class distribution of the leak-safe blended benchmark before train-only augmentation.

Dataset Preparation Pipeline

- **Deduplication:** removed redundant images to minimize leakage risk.
- **Standardization:** resized all fundus images to 300×300 .
- **Fundus preprocessing:** retinal FOV isolation with illumination/contrast normalization.
- **Train-only augmentation:** balanced training data without altering validation/test sets.

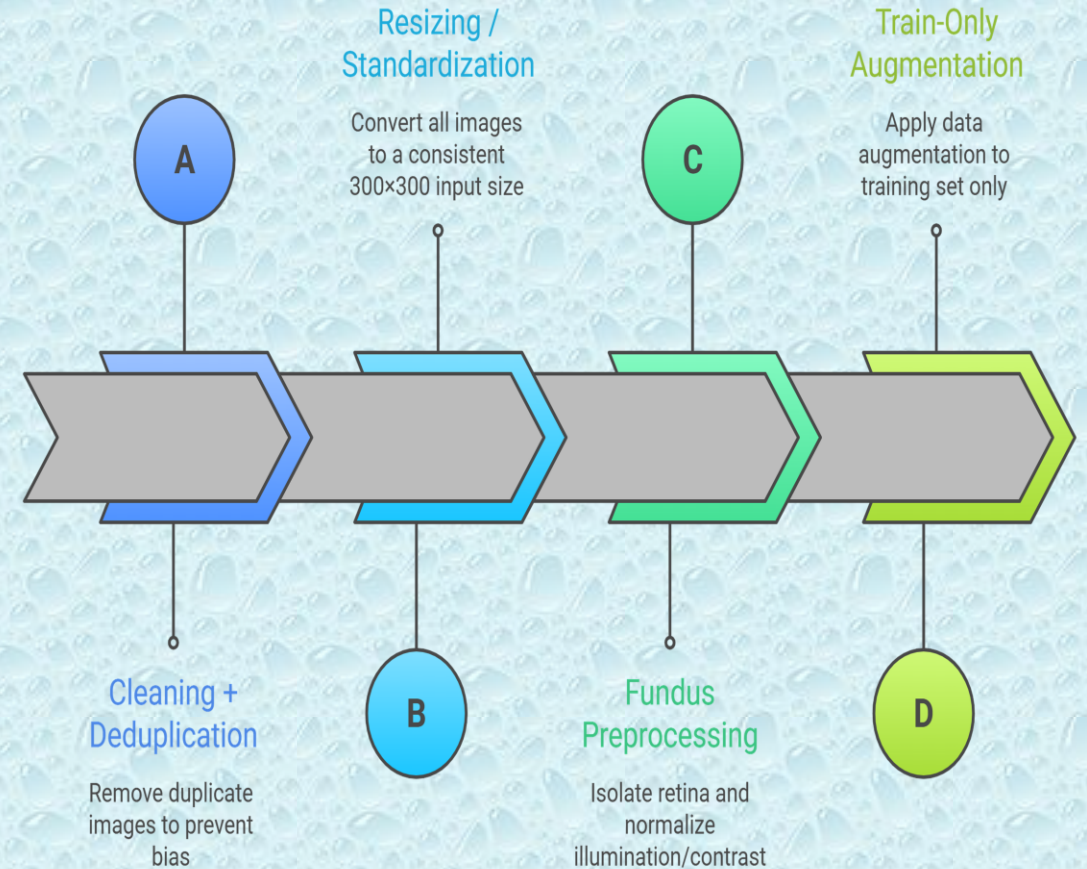


Figure 4: Dataset preparation workflow used before model training and evaluation.

Train-Only Balancing and Evaluation Splits

- ❖ **Training split balanced:** each DR grade increased to **945 images**.
- ❖ **Validation/Test unchanged:** no augmentation was applied to evaluation data.
- ❖ **Purpose:** reduce class imbalance during training while keeping evaluation realistic and unbiased.

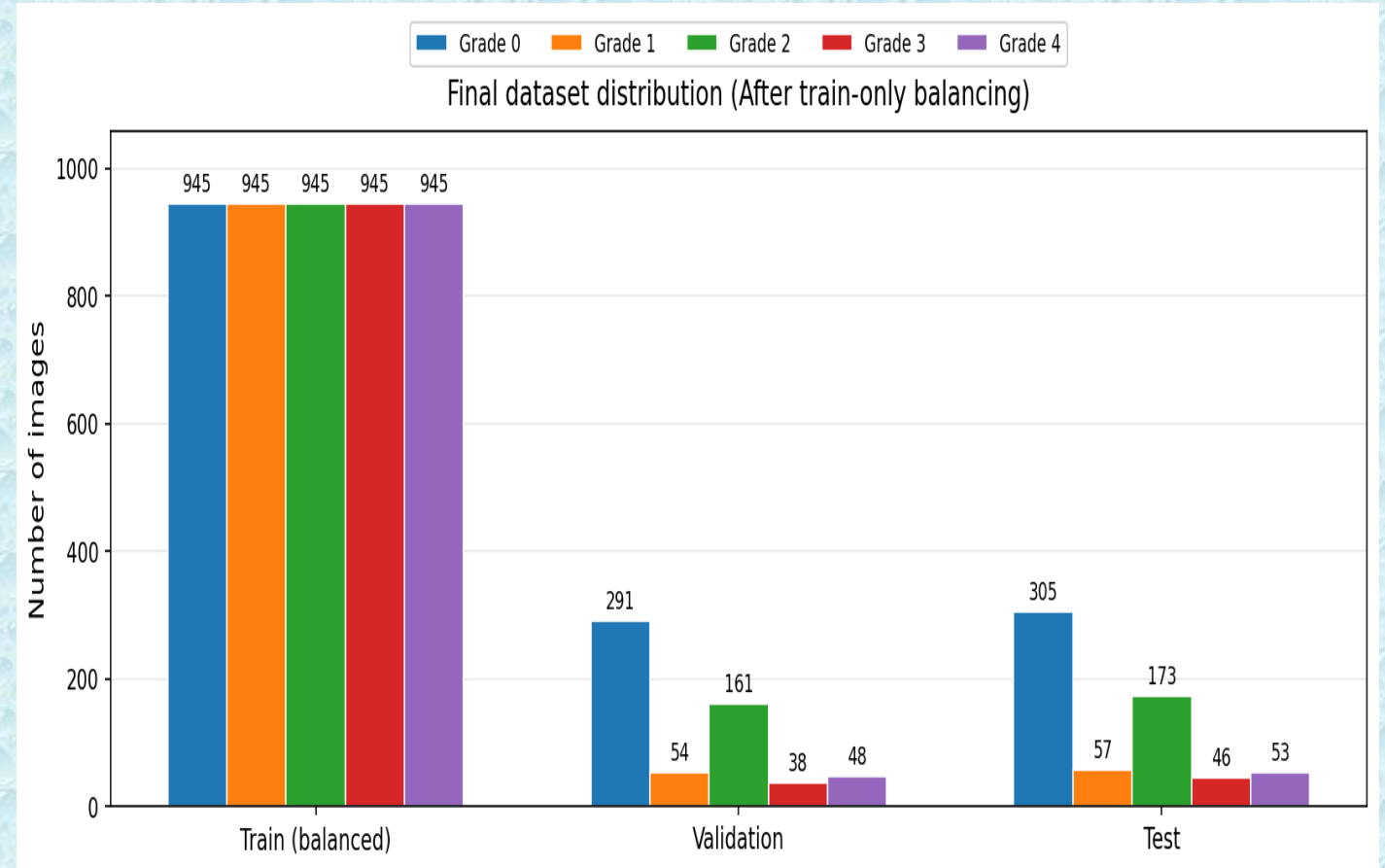


Figure 5: Split-wise class distribution after train-only balancing, with validation and test sets kept unchanged.

Proposed Stage-Wise Framework for Five- Class DR Grading

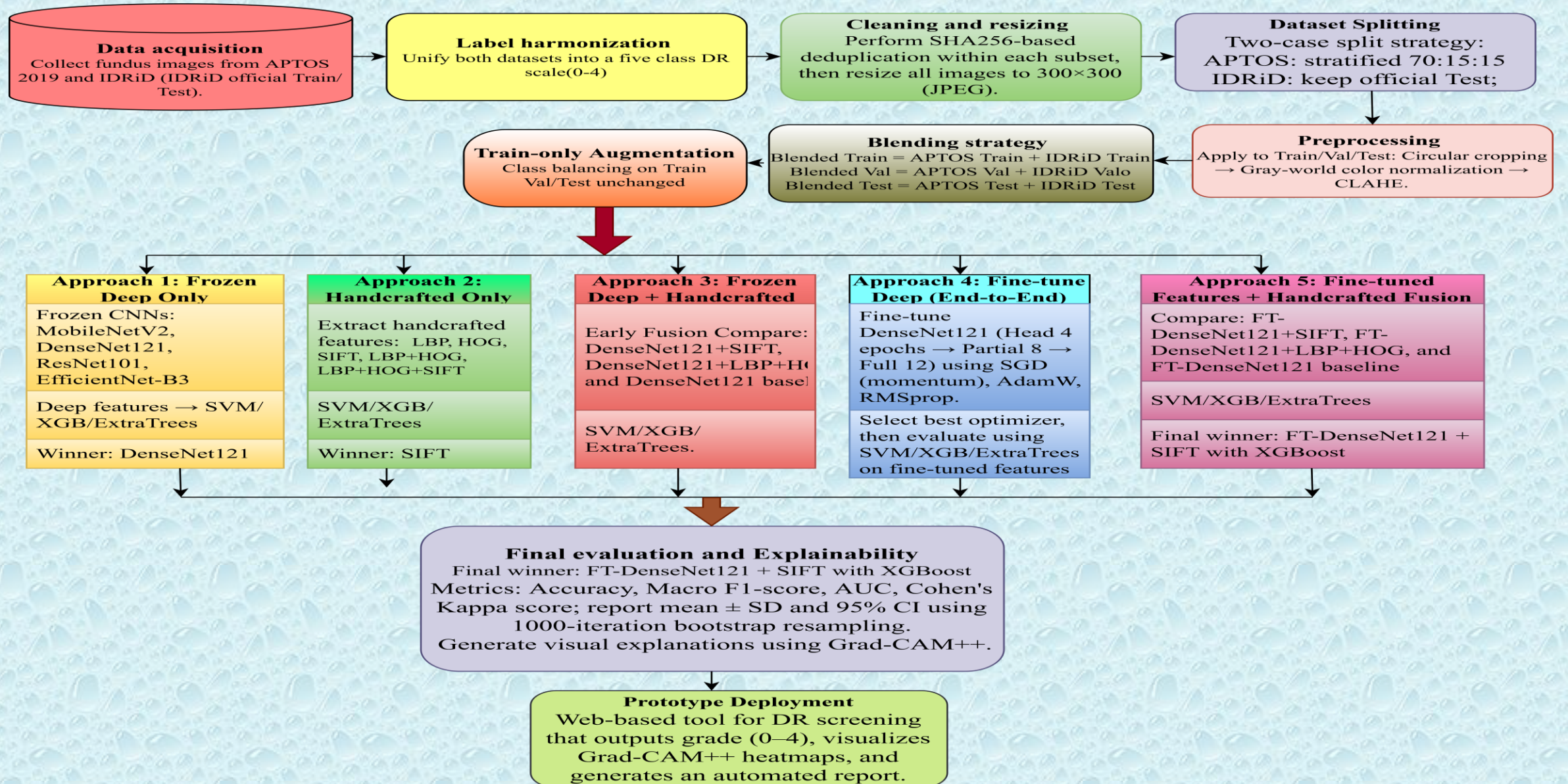


Figure 6: Overview of the proposed stage-wise framework for five-class DR grading.

Custom CNN Baselines: CNN-1 (Stacked CNN)

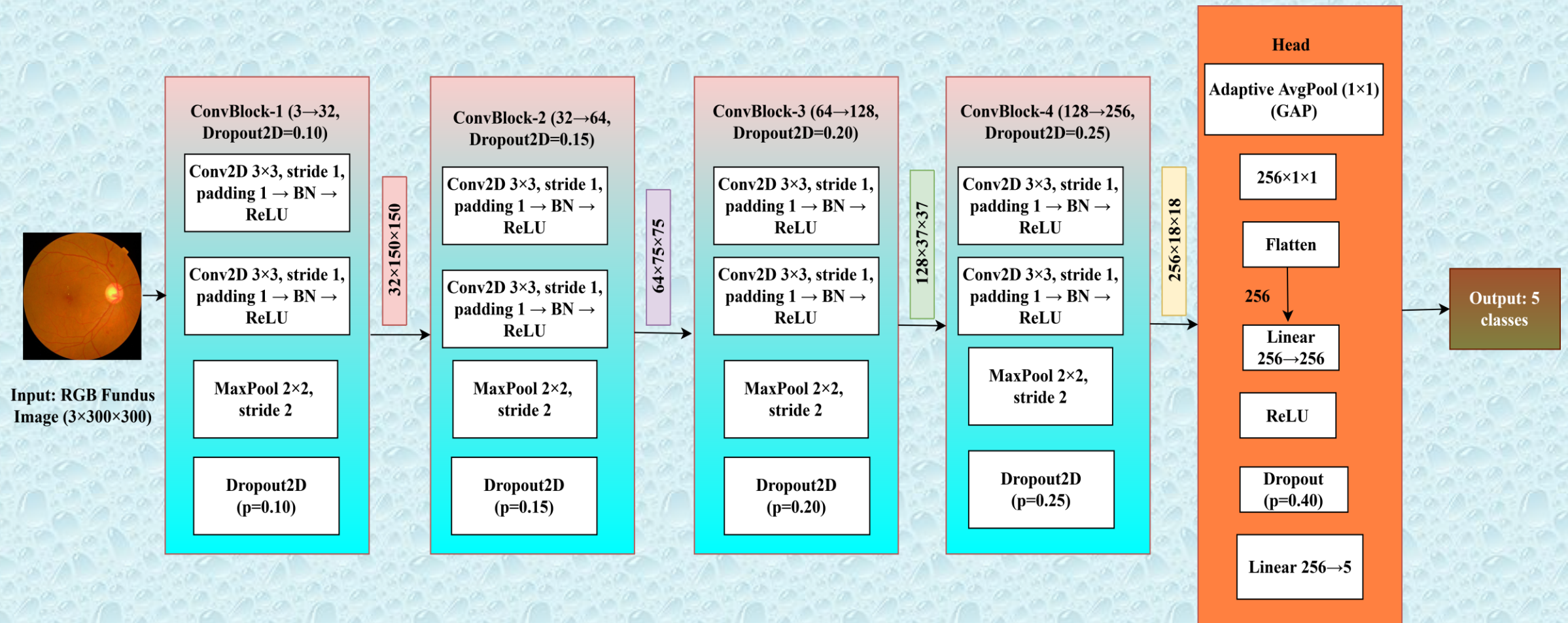


Figure 7: Architecture of CNN-1 (4×ConvBlock + GAP + FC head) for five-class DR grading.

CNN-2 (ResNet-Style with Squeeze-and-Excitation)

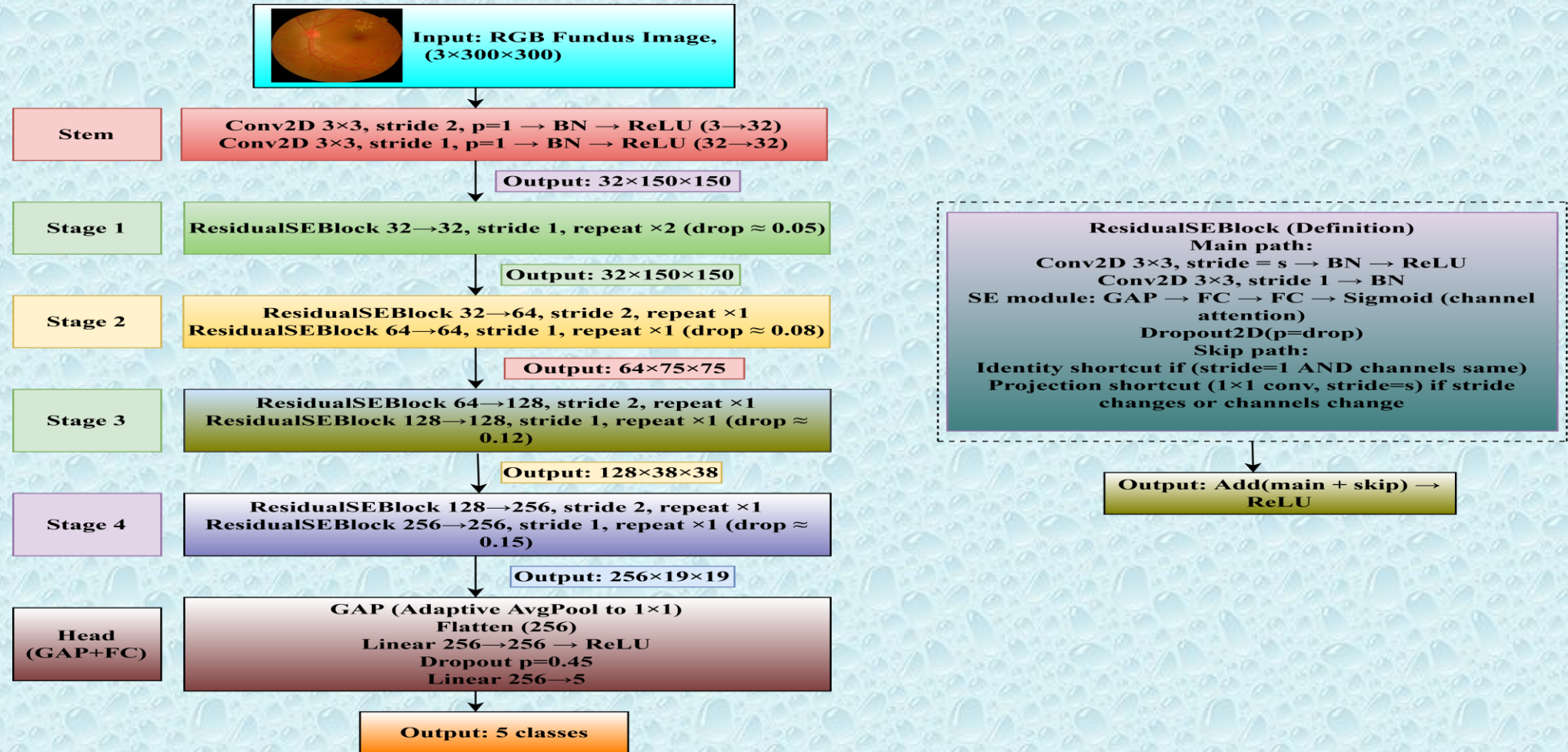


Figure 8: Architecture of CNN-2 (ResNet-style with SE blocks and projection shortcuts) for five-class DR grading.

CNN-3 (EfficientNet-lite MBConv)

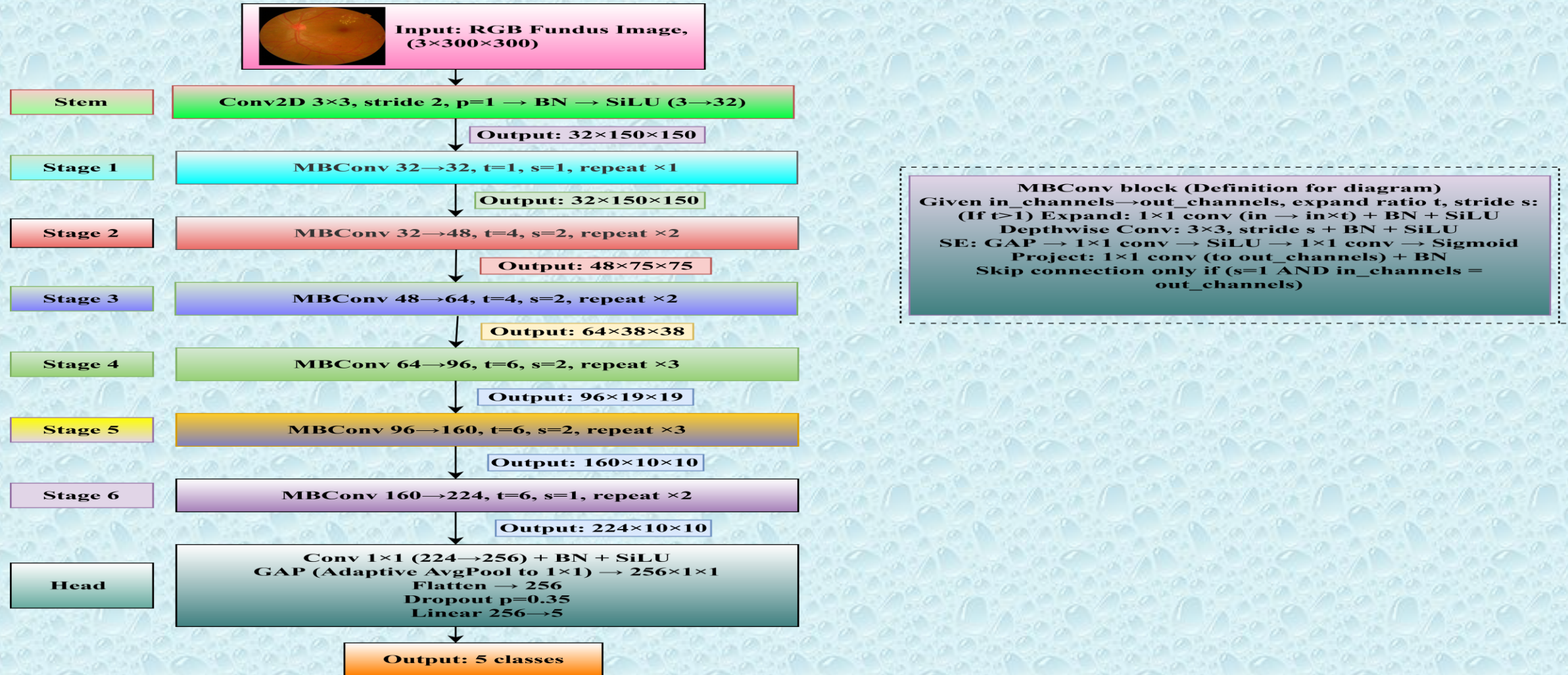


Figure 9: Architecture of CNN-3 (EfficientNet-lite MBConv with SE and DropPath) for five-class DR grading.

Evaluation Protocol and Final Model Reporting

- ❑ **Stage-wise selection:** Stages 1–4 used validation performance for component selection.
- ❑ **Locked final stage:** Stage-5 fusion candidates were fixed before held-out test evaluation.
- ❑ **Final reporting:** best locked Stage-5 test result, with no further tuning after test evaluation.
- ❑ **Metrics:** Accuracy, Macro-F1, Cohen's κ , OvR AUC, and confusion matrix.
- ❑ **Bootstrap uncertainty:** mean $\pm$ SD and 95% CI from 1,000 bootstrap resamples.

Implementation Environment

Hardware: Intel
Core i9-13900K
CPU, 64 GB
RAM, NVIDIA
RTX 4090 GPU

**Software
environment:**
Windows 11,
Python 3.12.3

**DL libraries &
GPU
acceleration:**
PyTorch 2.5.1,
torchvision
0.20.1, CUDA
12.1, cuDNN 9.1

**ML/CV
packages:**
scikit-learn
1.7.2, XGBoost
2.1.3, OpenCV
4.12.0

Final Stage-5 Test Performance

Feature setting (Stage-5)	Classifier	Accuracy	Macro-F1	Cohen's κ	AUC (OvR)
FT- DenseNet121 + SIFT-BoVW	SVM	77.60	62.75	66.19	94.13
	Extra Trees	77.13	63.25	65.27	94.53
	XGBoost	78.71	64.24	67.89	94.78
FT- DenseNet121 + (LBP+HOG)	SVM	75.39	59.47	62.36	93.24
	Extra Trees	72.71	53.23	57.80	90.90
	XGBoost	73.82	57.64	59.99	92.02
FT- DenseNet121 only	SVM	77.76	63.81	66.49	93.86
	Extra Trees	76.81	62.63	64.86	94.21
	XGBoost	77.92	63.63	66.73	94.45

Error Analysis: Confusion Matrix and ROC

Key takeaway: The final Stage-5 model shows strong overall class separability, but most misclassifications occur between clinically adjacent DR grades, especially **Moderate**, **Severe**, and **PDR**.

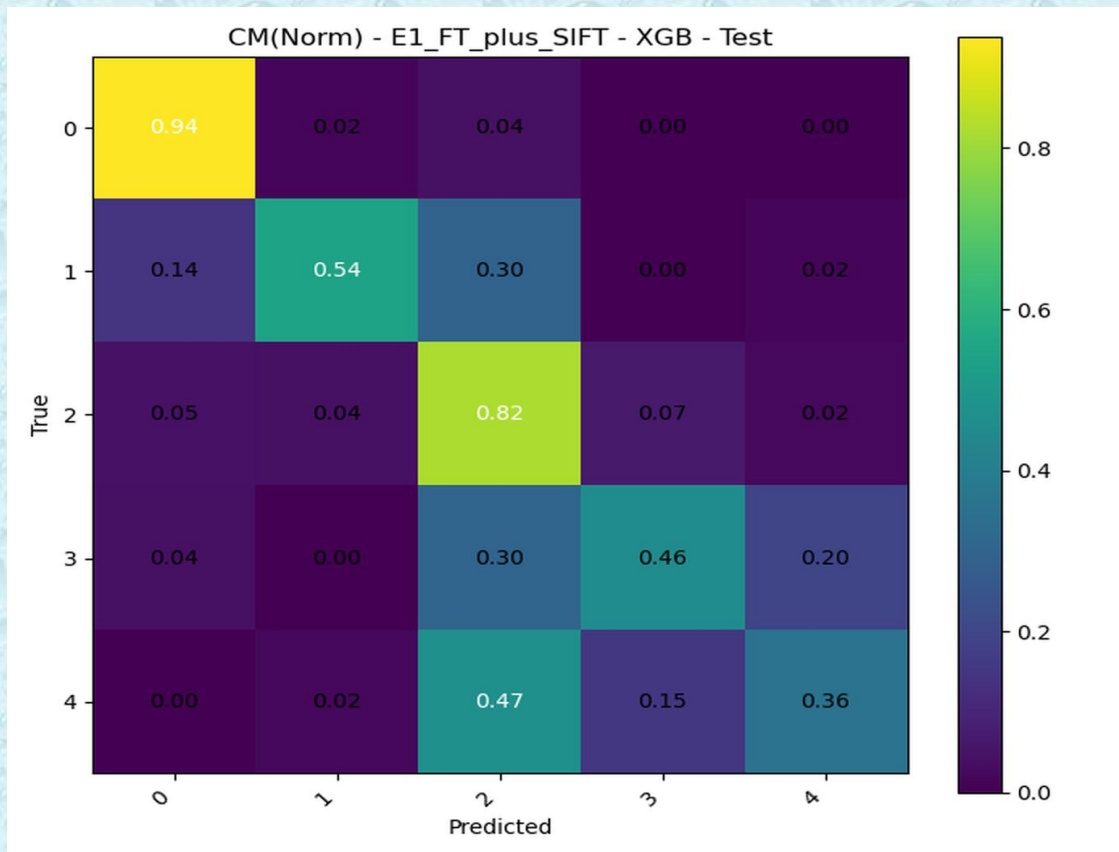


Figure 10: Normalized confusion matrix of the final Stage-5 model on the held-out test set.

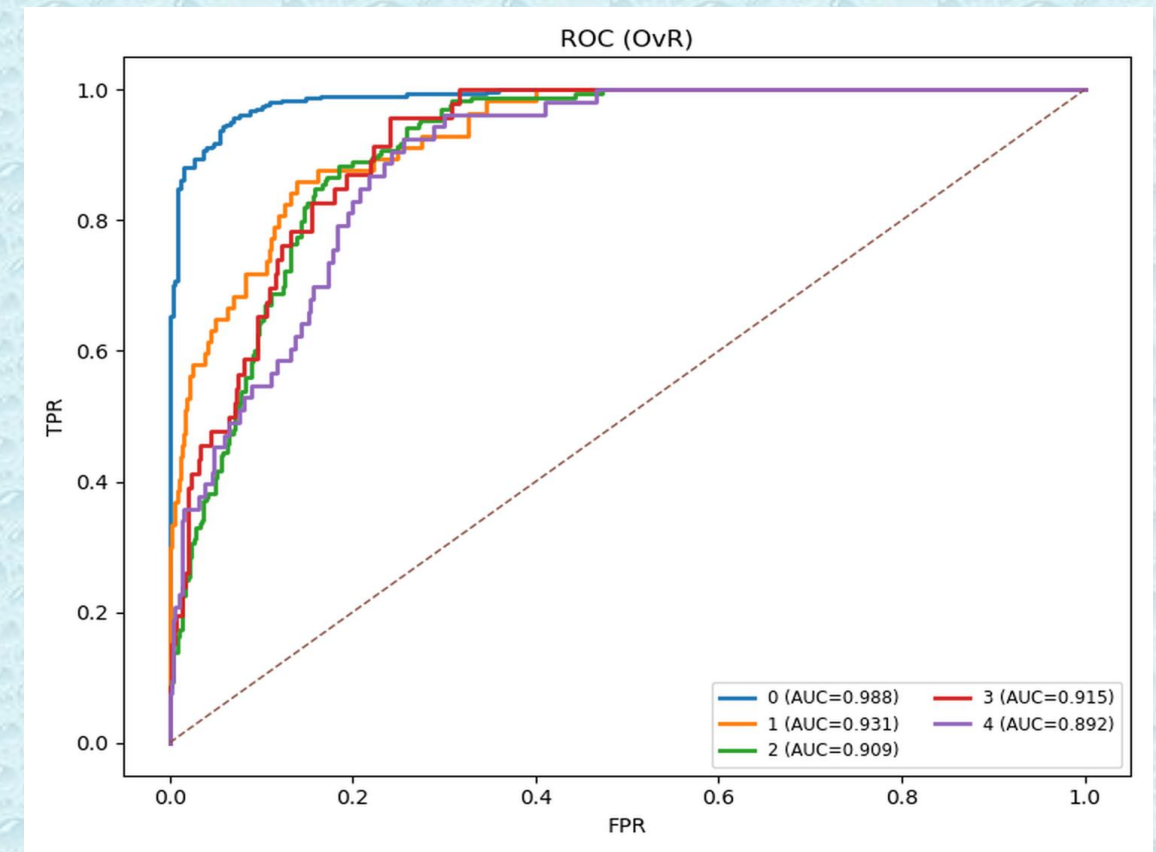
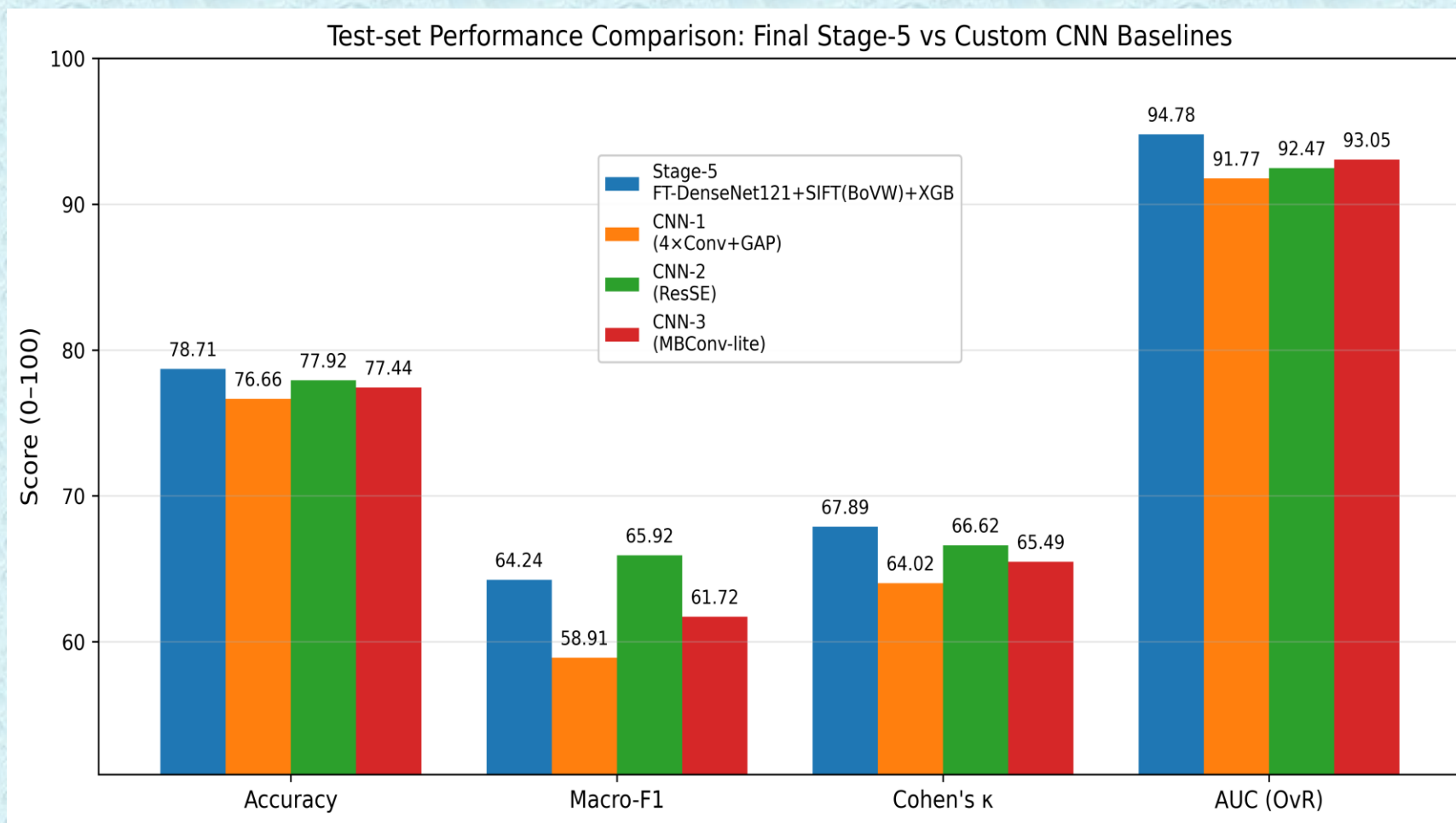


Figure 11: One-vs-rest ROC curves of the final Stage-5 model on the held-out test set.

Test-Set Comparison with Custom CNN Baselines

Method (Test)	Accuracy	Macro-F1	Cohen's κ	AUC (OvR)
Stage-5 (Final): FT-DenseNet121 + SIFT-BoVW + XGBoost	78.71	64.24	67.89	94.78
CNN-1 (4×ConvBlock + GAP + FC head)	76.66	58.91	64.02	91.77
CNN-2 (ResNet- style + SE + projection)	77.92	65.92	66.62	92.47
CNN-3 (MBConv / EfficientNet-lite style)	77.44	61.72	65.49	93.05

Metric-Wise Performance Comparison



Key takeaway: The final Stage-5 fusion model achieved the best Accuracy, Cohen's κ , and OvR AUC, while CNN-2 achieved the highest Macro-F1.

Figure 12: Metric-wise test-set performance comparison between the final Stage-5 model and custom CNN baselines.

Bootstrap Uncertainty Analysis

Method	Accuracy (mean±SD) [95% CI]	Macro-F1 (mean±SD) [95% CI]	Cohen's κ (mean±SD) [95% CI]	AUC (OvR) (mean±SD) [95% CI]
Stage-5 (FT-DenseNet121 + SIFT-BoVW + XGB)	78.71±1.65 [75.39, 81.86]	64.13±2.55 [59.07, 69.20]	67.88±2.30 [63.21, 72.36]	94.80±0.59 [93.54, 95.97]
CNN-1	76.66±1.68 [73.34, 79.97]	58.81±2.49 [54.03, 63.65]	64.01±2.33 [59.50, 68.54]	91.75±0.75 [90.26, 93.15]
CNN-2	77.90±1.71 [74.29, 80.91]	65.78±2.57 [60.91, 70.80]	66.56±2.44 [61.67, 70.94]	92.46±0.80 [90.86, 93.99]
CNN-3	77.45±1.67 [74.13, 80.60]	61.54±2.51 [56.74, 66.33]	65.47±2.33 [60.98, 69.82]	93.03±0.72 [91.69, 94.39]

Explainable Web-Based Screening Prototype



DIABETIC RETINOPATHY SCREENING TOOL
Powered by DSLR Lab, Pabna University of Science and Technology (PUST)
Developed by Md. Razu Ahmed, Department of Statistics, PUST
Under supervision of Prof. Dr. Md. Shamim Reza, Department of Statistics, PUST
"Diabetic Retinopathy doesn't wait – neither do we."

Backend: PUST-XDRFusionNet (Stage-5 XGB + DenseNet Grad-CAM++)
Device: cpu
Prototype: v1.0

Patient Information & Image Upload

Age (years)
e.g. 55

Gender
Select

History of diabetes
Select

Duration of diabetes (years)
e.g. 0.5, 5, 12.5

Eye
Select

Hypertension
Select

Upload a retinal fundus image
Accepted formats: .jpg, .jpeg, .png

Choose fileNo file chosen

Your image is processed locally using the fine-tuned model. This is a research prototype and does not store personal data externally.

Analyze imageTry sample image

Session History

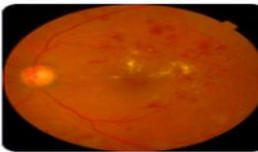
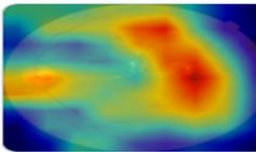
ID	Time	Age	Eye	DR Grade	Risk	Quality
DR-0001	2026-01-19 06:55	32	Right	4 – PDR	Urgent risk (PDR)	Good

Learn more about Diabetic Retinopathy

- Diabetic retinopathy (DR) is a microvascular complication of diabetes affecting the retina.
- Early stages may be asymptomatic but can progress to sight-threatening disease.
- Strict control of blood glucose, blood pressure and lipids reduces the risk of DR progression.
- Regular retinal screening and timely treatment (e.g. laser, intravitreal therapy) can prevent visual loss.

Important: This prototype is for screening support only and does not replace a comprehensive examination by a qualified ophthalmologist.

Screening Result (Prototype)

Uploaded image & Grad-CAM++
Original fundus image

Grad-CAM++ focus map

Estimated image quality: Good

Model screening summary

Record ID: DR-0001
Age / Eye: 32 years, Right eye
DR Grade: 4 – PDR
DR likelihood: 86.1%
Risk level: **Urgent risk (PDR)**

Explainable AI interpretation

Grad-CAM++ highlights regions that influenced the model decision. Use this as a visual aid only; it is not a clinical diagnosis.

Screening note

The pattern is consistent with PDR. If symptoms exist or quality is poor, consider retaking image.

Suggested action

If referable or urgent risk is shown, recommend an ophthalmologist/retina specialist review.

Class-wise model probabilities

Grade	Class name	Probability
0	No DR	0.0%
1	Mild NPDR	2.8%
2	Moderate NPDR	2.5%
3	Severe NPDR	8.5%
4	PDR	86.1%

Download Screening Report (PDF)

© 2026 Md. Razu Ahmed, DSLR Lab, Pabna University of Science and Technology (PUST). This is a research prototype.

Figure 13: Web-based screening prototype showing image upload, predicted grade, class probabilities, and Grad-CAM++ visualization.

Conclusion

- Developed a **leak-safe cross-source benchmark** using APTOS 2019 + IDRiD for five-class DR grading.
- Proposed a **stage-wise deep-handcrafted framework** for controlled comparison of deep, handcrafted, fusion, and fine-tuned representations.
- Identified **DenseNet121** and **SIFT-BoVW** as the strongest deep and handcrafted components.
- Final model: **FT-DenseNet121 + SIFT-BoVW + XGBoost**.
- Achieved strong overall test performance with **78.71% Accuracy**, **67.88% Cohen's κ** , and **94.80% OvR AUC**.
- Integrated the selected model into an explainable **Grad-CAM++ web prototype** for screening support.

Limitations

- ❑ Mixed-source evaluation improves diversity but does not replace external multi-center validation.
- ❑ Adjacent DR grades remain difficult due to subtle lesion overlap and class imbalance.
- ❑ Grad-CAM++ provides visual explanation but not lesion-level clinical verification.
- ❑ The web prototype is for screening-oriented research support, not standalone diagnosis.

Future Work

- ❖ Validate on larger multi-center and device-diverse datasets.
- ❖ Improve robustness under domain shift and poor image quality.
- ❖ Explore ordinal, cost-sensitive, and lesion-aware learning.
- ❖ Add confidence-aware referral flags and clinician feedback.
- ❖ Extend toward mobile screening workflows.

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Thank
You!

Q&A

**I would be
happy to
take any
questions.**





The End!